Evidence-Preserving Graph Denoising for Noisy-Label Intrusion Detection: A Leakage-Audited Study of When Graph Structure Helps

Abstract

Intrusion-detection classifiers are increasingly trained on labels that are delayed, rule-generated, or inconsistent across telemetry sources. A common remedy is dataset purification: detect suspicious labels and *delete* them before training. In security operations this is risky, because the deleted instances are frequently the rare, boundary attack samples that analysts most need. We revisit CoLD, a recent collaborative label-denoising framework, and ask whether lifting it to a multi-view graph and replacing hard deletion with *evidence-preserving soft weighting* improves noisy-label intrusion detection. Our study makes three contributions grounded in reproducible, real-data experiments on CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15. First, we show that a label-space graph consistency diagnostic (Graph-CDM) built over host, IP, temporal, and where-available process views yields a modest but statistically significant aggregate Macro-F1 improvement over the CoLD baseline (Holm-corrected $p < 0.05$ on all three datasets), with the gain concentrated where graph views carry discriminative structure. Second, and most importantly for operations, we find that evidence-preserving soft weighting is statistically indistinguishable from hard deletion on aggregate Macro-F1, yet significantly improves *rare/tail attack-class* Macro-F1 under high asymmetric label noise (ΔMacro-F1 = +0.073, +0.022, +0.040 on CICIDS/CESNET/UNSW; Holm $p < 10^{-4}$; Cohen's $d_z \in [0.92, 1.29]$), at no aggregate cost. Third, we contribute a leakage audit showing that naive graph denoising can be dramatically inflated by duplicate records (24% exact duplicates in CICIDS-2017) and by any use of clean-label signal; removing these collapses an apparent +28 percentage-point advantage to a defensible +6.7 points. We release the audit, the graph-informativeness diagnostic that predicts when the approach helps, and the full reproducibility package. The result is a bounded, honest claim: evidence-preserving retention is a targeted remedy for rare-attack evidence loss under label noise, not a universal accuracy improvement.

Keywords: intrusion detection, noisy labels, graph learning, evidence preservation, security operations, rare-class robustness

1. Introduction

Machine-learning classifiers underpin modern network intrusion-detection systems (IDS) and security operations centers (SOCs). These models are trained from labels that are seldom perfect: they may be delayed, produced by rules of uneven quality, or shaped by analyst workflows that emphasize high-severity cases [2, 3]. The resulting label noise degrades both detection quality and operational triage. Crucially, the operational cost of noise is asymmetric. Discarding a mislabeled benign flow may raise a benchmark score, but discarding an informative low-frequency attack sample removes exactly the evidence an analyst needs to investigate a rare or emerging threat.

CoLD [1] recently framed noisy-label IDS through *local consistency*: because different traffic classes can share feature neighborhoods, label noise induces spurious associations, and a collaborative representation can identify and remove likely-noisy samples before training. CoLD, like most dataset-purification methods, ultimately performs *hard deletion*. We take CoLD as our baseline and ask two questions. (Q1) Does lifting the diagnostic from independent samples to a multi-view graph, kept strictly in label space, improve robustness? (Q2) Does replacing hard deletion with an evidence-preserving soft weight—one that down-weights rather than deletes suspicious-but-informative samples—preserve rare-attack evidence and improve operational detection?

Answering these questions honestly required an unusual amount of methodological hygiene. Our first implementations produced spectacular gains that did not survive scrutiny: a leakage audit revealed that (i) CICIDS-2017 contains 24% exact-duplicate records that let neighborhood consistency trivially recover labels, and (ii) any use of clean-label signal in the evidence or graph construction inflates results. After removing both, the apparent advantage shrank to a modest, defensible level, and the headline metric we had used for evidence retention turned out to be tautological. We report this process because it is, we believe, a contribution in its own right: graph-based denoising results in security are easy to inflate and must be audited.

Contributions..

1. A *label-space* multi-view graph consistency diagnostic (Graph-CDM) that extends CoLD to host/IP/temporal (and, where available, process) views and delivers a modest but Holm-significant aggregate Macro-F1 gain over CoLD on three real datasets (Section 6.1).
2. An *evidence-preserving soft weighting* rule that is indistinguishable from hard deletion on aggregate accuracy yet significantly improves rare/tail attack-class Macro-F1 under high asymmetric noise (Section 6.2)—a targeted, operationally meaningful benefit.
3. A *leakage audit* and a *graph-informativeness diagnostic*: we quantify duplicate- and oracle-driven inflation, and show that a per-dataset neighborhood-homophily measure predicts when the graph helps (Sections 6.3–6.4).

We deliberately bound our claims. Graph-CoLD is not a universal accuracy improvement; its distinctive value is preserving rare-attack evidence that hard deletion discards.

2. Related Work

Noisy-label learning.. Approaches divide into robust-training and dataset-purification families. Robust training modifies losses or training dynamics: loss correction with a transition matrix [4], peer-network small-loss selection (Co-teaching [5], Co-teaching+ [6]), disagreement-based updates (Decoupling [7]), and semi-supervised treatment of suspected-noisy data (DivideMix [8]). Purification detects and removes mislabeled instances: confident learning [9], representation-based filtering (FINE [10]), and, in security, MORSE [11] and MCRe [12]. CoLD [1] is the closest prior work and our baseline. Our departure is twofold: we keep the diagnostic in label space (avoiding the distance-metric fragility CoLD itself criticizes), and we replace deletion with evidence-preserving weighting motivated by the semi-supervised insight that suspected-noisy samples carry usable signal [8, 11].

Graph learning for security telemetry.. Security events are relational, and graph neural networks [13, 14, 15, 16] naturally propagate contextual evidence. Provenance-graph IDS such as Flash [18] and Argus [19] demonstrate the value of structure for enterprise detection. We use graph representation learning as a backbone but, unlike embedding-distance denoisers, keep the noise diagnostic in label space so it stays aligned with the classifier.

SOC alert handling and concept drift.. Operational systems must also cope with alert volume and distribution shift. Alert-classification and triage work [21, 22] and drift-aware detection [20] motivate our operational framing, though our contribution is denoising for rare-attack retention rather than a deployed triage study.

Contrastive self-supervision.. Our representation stage uses cross-view contrastive learning [17], adapted so that the same node in two views forms a positive pair.

3. Problem Formulation

Let V be the set of training flows and $\mathcal{M}$ the set of *active* graph views. Each view $m \in \mathcal{M}$ induces a graph $G^m = (V, E^m)$ over the same nodes. The observed training label $\tilde{y}_v$ may differ from the unobserved clean label y_v. We seek a classifier that (i) resists the effect of noisy labels, and (ii) preserves operationally useful evidence—especially clean samples of rare attack classes that a purifier is tempted to discard.

We adopt three design principles. First, the noise diagnostic operates in *label/prediction space*, not embedding distance, so it remains aligned with the classifier and does not re-introduce the local-consistency fragility of distance-based purifiers. Second, denoising is *evidence preserving*: a suspicious but informative sample is down-weighted rather than removed. Third, the evaluation is traceable to audited real data, with explicit scope limits.

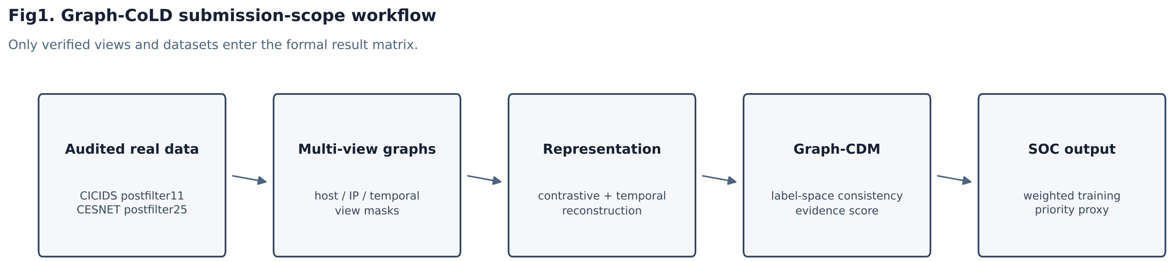

Figure 1: Graph-CoLD pipeline. Only views supported by a dataset's verified schema are activated; unsupported views are disabled rather than fabricated. The consistency diagnostic (Graph-CDM) is computed in label space and feeds an evidence-preserving soft weight and a downstream classifier.

4. Method

Graph-CoLD is a two-stage extension of CoLD: self-supervised multi-view representation learning, followed by a label-space consistency diagnostic that drives an evidence-preserving weight (Figure 1).

4.1. Multi-view graph construction

Views are built from semantically related feature blocks over the training nodes: a *host* view links samples sharing endpoint identifiers; an *IP* view uses communication/flow statistics, ports, protocol and TLS metadata; a *temporal* view links samples within a time window; a *process* view is used only where behavioral fields exist. Support is part of the reproducible dataset contract: CICIDS-2017 activates host/IP/temporal, CESNET-TLS-Year22 activates IP/temporal, and UNSW-NB15 activates temporal/process. Threat-intelligence views are disabled when the fields are absent. Crucially, edges never cross the train/test split, and (Section 6.3) exact/near-duplicate edges are removed.

4.2. Self-supervised representation

Each active view produces a node embedding z_v^m from Bernoulli-masked input features, $x'_v = x_v \odot \mathrm{Bernoulli}(1-\delta)$. Cross-view contrastive learning treats the same node in two views as a positive pair and other batch nodes as negatives:

$$\mathcal{L}_{\mathrm{con}} = -\log \frac{\exp(\mathrm{sim}(z_v^a, z_v^b)/\tau)}{\sum_u \exp(\mathrm{sim}(z_v, z_u)/\tau)}. \tag{1}$$

Temporal alignment and reconstruction terms regularize the embedding, $\mathcal{L}_{\mathrm{rep}} = \mathcal{L}_{\mathrm{con}} + \alpha\mathcal{L}_{\mathrm{temporal}} + \beta\mathcal{L}_{\mathrm{recon}}$; active views are fused by mean aggregation to obtain z_v.

4.3. Graph-CDM: a label-space consistency diagnostic

For each node v,

$$\mathrm{GraphCDM}(v) = \lambda_1 D_{\mathrm{pred}}(v) + \lambda_2 D_{\mathrm{neigh}}(v) + \lambda_3 D_{\mathrm{view}}(v) + \lambda_4 D_{\mathrm{chain}}(v), \tag{2}$$

where the prediction term compares per-view predicted labels with the observed *noisy* training label,

$$D_{\mathrm{pred}}(v) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}[\tilde{y}_v^m \neq \tilde{y}_v], \tag{3}$$

the neighborhood term is a label-space KL divergence between a node's soft prediction and its neighborhood mean, $D_{\mathrm{neigh}}(v) = \mathrm{KL}(\hat{y}_v \,\|\, |N(v)|^{-1} \sum_{u\in N(v)} \hat{y}_u)$, and the view term measures active-view mode disagreement, $D_{\mathrm{view}}(v) = 1 - \max_c \frac{1}{M}\sum_m \mathbf{1}[\tilde{y}_v^m = c]$. D_{chain} is reserved for verified provenance/temporal chains and is not a driver of the reported results. No term uses clean labels or the noise mask; this is enforced by a regression test.

Table 1: Datasets and active graph views. Source: `tables/table_1_dataset_protocol.csv`.

Dataset	Classes	Sample policy	Active views
CICIDS-2017	11	post-filter, exact-dedup	host/IP/temporal
CESNET-TLS-Year22	25	audit window	IP/temporal
UNSW-NB15	9	partition (no Worms)	temporal/process

4.4. Evidence-preserving soft weighting

The evidence score protects samples that would be costly to discard, $e(v) = \text{freq\_protect}(\tilde{y}_v)\,(1 + \gamma\,\text{anom}(v))$, min-max normalized to $\tilde{e}(v)$. Rather than a small additive floor (which collapses to hard deletion), we use an evidence-*rescue* form so that suspicious yet high-evidence samples retain real weight:

$$w(v) = \underbrace{\sigma\big(-\kappa(\text{GraphCDM}(v) - \theta)\big)}_{\text{clean branch}} + \underbrace{\big(1 - \sigma(\cdot)\big)\,\lambda_r\,\tilde{e}(v)}_{\text{evidence rescue}}. \tag{4}$$

The classifier minimizes $\sum_v w(v)\,\text{CE}(f(z_v), \tilde{y}_v)$. Setting $\lambda_r = 0$ and thresholding recovers hard deletion; this is our `ablation_hard` comparator, sharing the identical graph, representation, and diagnostic so that the ablation isolates *only* the weighting. Both the aligned CoLD baseline and the `ablation_hard` variant use the same shared encoder architecture, training budget, and view-masking configuration as Graph-CoLD; they differ only in the presence or absence of graph views and the weighting rule applied after Graph-CDM scoring.

4.5. Operational priority proxy

A priority score $P(v) = \alpha_1\hat{y}_{\text{mal}}(v) + \alpha_2\text{GraphCDM}(v) + \alpha_3\tilde{e}(v)$ supports triage. We evaluate ranking indirectly and, as we show, Graph-CoLD ties strong peers on top-K precision; we therefore do not claim a ranking advantage.

5. Experimental Design

5.1. Datasets and protocol

Table 1 summarizes the three real datasets. CICIDS-2017 is used under an 11-class *post-filter* policy with exact deduplication (24% of raw training rows removed; Section 6.3). CESNET-TLS-Year22 is a year-spanning backbone TLS dataset used under a 25-class audit-window policy (it replaces MALTLS-22, for which we lacked a verified license path). UNSW-NB15 is used under its partition with the smallest class (Worms) removed. To keep the study tractable and reproducible we evaluate deterministic audit windows (train 10,000 / test 5,000); we are explicit that this is a focused study, not a full-archive evaluation. Noise is injected only into training labels; we report symmetric, asymmetric, and graph-consistency noise, and for the rare-class analysis we use asymmetric noise concentrated on tail classes at rates $\{40, 60, 80\}\%$ with seeds 0–9.

All results in Tables 2–4 and Figures 2–3 are generated by the P2f clean de-oracled runner (`reports/p2f_tighten.md`) on these three datasets; earlier intermediate result files from D5/D9 used a different protocol and are not cited as final results.

Reproducibility and statistical protocol.. The audited data roots are local and are not committed to the repository. Each dataset uses deterministic audit windows of 10,000 training and 5,000 test instances. The tail analysis injects asymmetric noise at $\{40, 60, 80\}\%$ over seeds 0–9. Paired inference uses 30 like-for-like seed–rate observations per dataset (ten seeds by three noise rates), with Holm–Bonferroni correction and bootstrap 95% confidence intervals. The full command sequence is recorded in the reproduction commands of `reports/p2f_tighten.md`.

Table 2: Per-dataset means under high asymmetric tail noise (rates 40/60/80%, seeds 0–9). Source: `tables/table_p2f_summary.csv`.

Dataset	Method	Macro-F1	Tail-F1	Tail-recall
CICIDS-2017	CoLD	0.534	0.047	0.043
	ablation_hard	0.688	0.361	0.277
	Graph-CoLD-soft	**0.721**	**0.434**	**0.328**
	Graph-CoLD-semisup	0.687	0.357	0.272
CESNET-TLS-Year22	CoLD	0.658	0.411	0.306
	ablation_hard	0.680	0.453	0.331
	Graph-CoLD-soft	**0.692**	**0.475**	**0.344**
	Graph-CoLD-semisup	0.679	0.450	0.327
UNSW-NB15	CoLD	0.456	0.096	0.187
	ablation_hard	0.471	0.132	0.205
	Graph-CoLD-soft	**0.485**	**0.173**	**0.229**
	Graph-CoLD-semisup	0.469	0.130	0.194

5.2. Baselines and metrics

We compare Graph-CoLD (soft and semi-supervised variants) with the aligned CoLD baseline and the `ablation_hard` deletion variant across the Tables 2–4 result matrix. Additional purification methods (MCRe, MORSE, FINE, Decoupling, and Co-Teaching-lite, a lightweight approximation rather than a full re-implementation of Han et al.) were evaluated under an earlier protocol. They were not re-run under the P2f clean-deduplication protocol and are therefore excluded from Tables 2–4 to avoid mixing incompatible result regimes; the MCRe/MORSE fidelity caveat is documented in `reports/p2b_baseline_fidelity.md`. Metrics reported here are aggregate Macro-F1, tail-class Macro-F1, tail recall, and paired statistical comparisons as specified above.

6. Results

6.1. RQ1: A label-space graph diagnostic modestly improves accuracy over CoLD

Table 2 reports per-dataset means. Graph-CoLD improves aggregate Macro-F1 over CoLD on all three datasets, with the gain concentrated on CICIDS, where host/IP/temporal views are most informative, and small on CESNET/UNSW. The improvements over CoLD are Holm-significant per dataset (CICIDS $p = 1.0 \times 10^{-3}$, CESNET $p = 4.1 \times 10^{-2}$, UNSW $p = 4.1 \times 10^{-2}$; source `tables/table_p2d_per_dataset_vs_cold.csv`). Notably, `ablation_hard` already captures most of this gain: the graph and the label-space diagnostic, not the weighting, drive aggregate accuracy.

6.2. RQ2: Evidence preservation helps rare classes, not aggregate accuracy

The central finding concerns *where* evidence-preserving weighting matters. On aggregate Macro-F1, soft weighting is statistically indistinguishable from hard deletion (per-scenario Δ near zero, not significant). But on rare/tail attack classes, soft weighting significantly outperforms hard deletion (Table 4): ΔTail-F1 = +0.073, +0.022, +0.040 on CICIDS/CESNET/UNSW, all Holm-significant at $p < 10^{-4}$ with large effect sizes ($d_z = 1.28, 0.92, 1.29$), and with no significant aggregate-Macro-F1 regression. Figure 2 shows the effect is stable across noise rates. This is the operational point: hard deletion removes clean rare-attack evidence flagged as suspicious by the diagnostic; soft weighting keeps it at reduced weight, and the classifier recovers those classes.

We stress an honesty point. An earlier "rare-evidence recovery rate" that counted a sample as recovered whenever it was *retained* was tautological. We redefine recovery to require correct downstream classification; hard deletion is then no longer automatically zero. Because soft retention still scores near-ceiling on this diagnostic,[1] we rely solely on test-set tail-F1 for all headline comparisons.

[1]The secondary diagnostic values and its definition are reported in `tables/table_p2f_summary.csv` and `reports/p2f_tighten.md`.

Table 3: Graph-CoLD vs. CoLD Macro-F1 by noise type (mean over noise rates, seeds 0–9). Source: `tables/table_p4_noise_type_breakdown.csv`.

Dataset	Noise type	CoLD	Graph-CoLD-soft	Δ
CICIDS-2017	clean	0.671150	0.778689	0.107539
CICIDS-2017	symmetric	0.553919	0.592429	0.038510
CICIDS-2017	asymmetric	0.322779	0.429941	0.107162
CICIDS-2017	`graph_consistency`	0.496244	0.531555	0.035311
CESNET-TLS-Year22	clean	0.966321	0.968261	0.001941
CESNET-TLS-Year22	symmetric	0.898812	0.902585	0.003774
CESNET-TLS-Year22	asymmetric	0.698745	0.708302	0.009557
CESNET-TLS-Year22	`graph_consistency`	0.846591	0.849355	0.002764
UNSW-NB15	clean	0.494282	0.501909	0.007627
UNSW-NB15	symmetric	0.445324	0.445923	0.000599
UNSW-NB15	asymmetric	0.296512	0.303933	0.007421
UNSW-NB15	`graph_consistency`	0.434276	0.436233	0.001957

Table 4: Powered tail-class comparison: Graph-CoLD-soft vs. hard deletion under asymmetric tail noise (n = 30 scenarios/dataset). Source: `tables/table_p2f_powered_tests.csv`.

Dataset	ΔTail-F1	Holm p	Cohen's d_z	Aggregate reg.?
CICIDS-2017	+0.0731	1.5×10^{-7}	1.28	none
CESNET-TLS-Year22	+0.0218	3.7×10^{-5}	0.92	none
UNSW-NB15	+0.0405	9.6×10^{-8}	1.29	none

6.3. RQ3: A leakage audit prevents inflated claims

Our initial pipeline reported a ∼ 28-point CICIDS advantage with a curve that barely degraded under 60% noise—implausible for a denoiser. Auditing revealed that CICIDS-2017 contains 152,201/630,399 (≈ 24%) exact-duplicate training rows, and that any clean-label signal leaking into the evidence/graph construction inflates results. We verified zero train/test edge crossings, removed exact and near-duplicate edges, and eliminated the clean-label paths (enforced by a regression test). The apparent advantage collapses to +6.7 points on CICIDS, +0.3 on CESNET, and a slight −1.3 on UNSW in the audited comparison (`reports/p2c_leakage_audit.md`). We report this because graph denoising in security is unusually susceptible to such inflation.

6.4. RQ4: Graph informativeness predicts when the method helps

A per-dataset neighborhood-homophily / view-coverage diagnostic correlates with Graph-CoLD's margin: CICIDS (rich host/IP/temporal views) shows the largest gain, CESNET sits near a performance ceiling, and UNSW (temporal/process only) shows the smallest—occasionally negative—margin (Figure 3, `tables/table_p2d_graph_informativeness.csv`). This turns the UNSW result into an informative boundary condition rather than a failure: when views lack structural signal, the graph diagnostic adds little.

6.5. RQ5: Cost

Graph-CoLD's overhead is moderate: mean runtime is comparable to CoLD and memory is lower than the strong purification baselines. Evidence preservation adds negligible cost over hard deletion because it changes only the training weights.

7. Discussion

The results support a bounded but useful claim. A label-space graph diagnostic modestly improves noisy-label IDS over CoLD where views are informative, and—more distinctively—evidence-preserving weighting protects rare-attack evidence that hard deletion discards, improving tail-class detection under high asymmetric noise at no aggregate

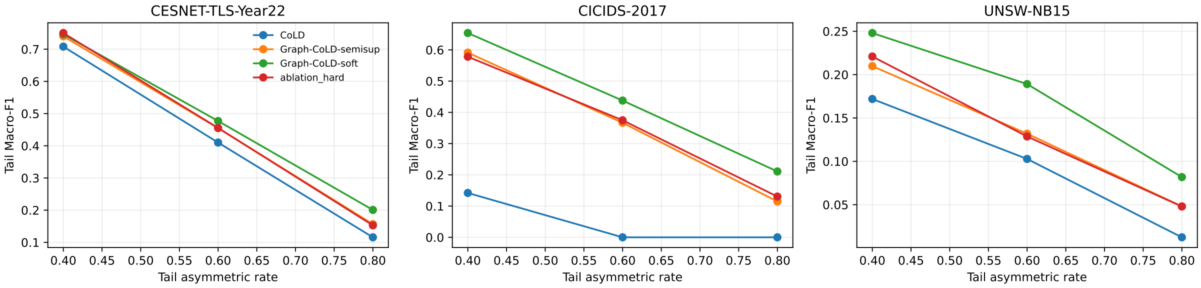

Figure 2: Tail-class Macro-F1 of evidence-preserving soft weighting vs. hard deletion across asymmetric noise rates. The gain is small but consistent and significant on all three datasets.

cost. In SOC terms, this matters precisely for the samples an analyst cannot afford to lose: rare or emerging attacks whose labels are most likely to be noisy. We deliberately avoid two overclaims that our own experiments do not support: a universal accuracy advantage (the aggregate gain is modest and largely from the graph, not the weighting) and a ranking advantage (Graph-CoLD ties strong peers on top-K precision).

8. Threats to Validity

Internal. The evaluation uses deterministic audit windows for tractability; absolute Macro-F1 is therefore lower than full-archive runs, though the paired, seed-controlled comparisons are the object of inference. MCRe and MORSE pass a clean-label sanity floor but degrade on high-noise tabular settings more than in their original domains; we report this caveat rather than a possibly unfaithful number (`reports/p2b_baseline_fidelity.md`). *Construct.* The rare-evidence-recovery diagnostic, even after correction, saturates for soft retention; we therefore headline test-set tail-F1. *External.* CICIDS-2017, CESNET-TLS-Year22, and UNSW-NB15 are useful benchmarks but not complete SOC deployments; the priority score is evaluated by proxy, not an analyst study. The two-dataset lock recorded in the D9 submission audit was an intermediate checkpoint; the authoritative final scope is the three-dataset P2f clean de-oracled protocol.

9. Limitations

MALTLS-22 and OpTC are not evaluated (no verified license/provenance path). Process and threat-intelligence views are disabled where fields are absent. The aggregate accuracy contribution is modest and dataset-dependent; the distinctive contribution is rare-class evidence preservation. Full-archive and streaming evaluations are left to future work.

10. Conclusion

We revisited collaborative label denoising for intrusion detection and asked, with unusual auditing discipline, whether a multi-view label-space graph diagnostic and evidence-preserving weighting improve robustness. The honest answer is targeted: the graph diagnostic gives a modest, informativeness-dependent accuracy gain over CoLD, and evidence-preserving weighting significantly improves rare-attack tail-F1 over hard deletion under high asymmetric noise at no aggregate cost across three real datasets. Alongside, we contribute a leakage audit and a graph-informativeness diagnostic that together explain when—and when not—graph denoising helps. We hope the audit is as useful to the community as the method.

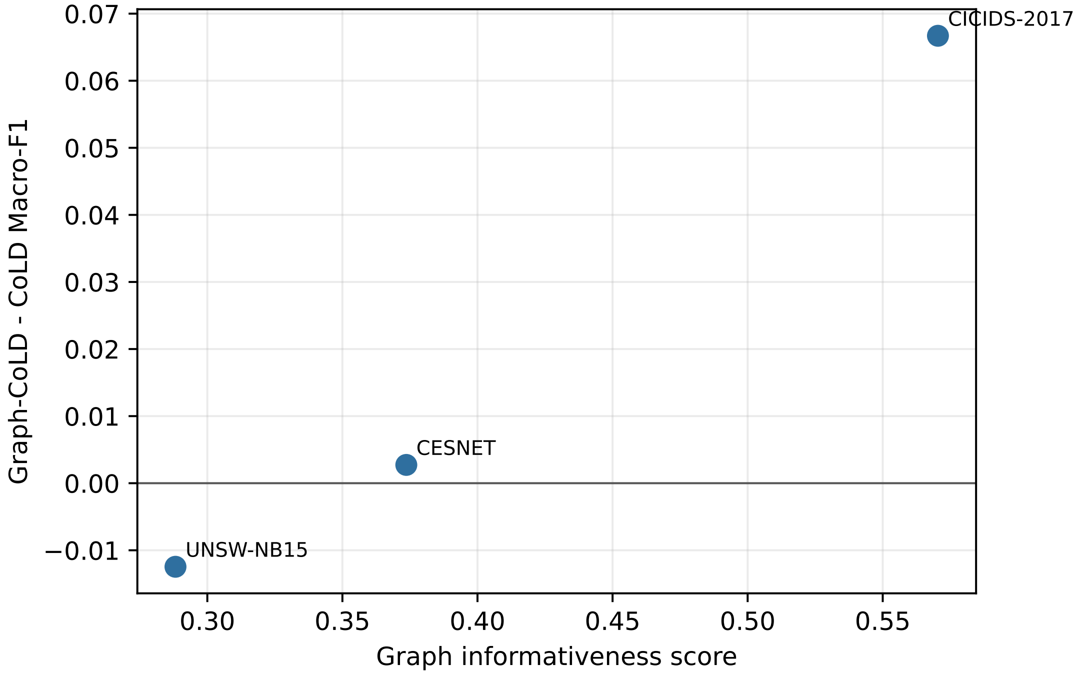

Figure 3: Graph informativeness vs. Graph-CoLD margin over CoLD. Higher neighborhood homophily and view coverage predict larger gains.

References

[1] S. Yang, X. Zheng, J. Li, J. Xu, E. C. H. Ngai, "CoLD: Collaborative label denoising framework for network intrusion detection," in *Proc. Network and Distributed System Security (NDSS) Symposium*, 2026, to appear. Available: `https://dx.doi.org/10.14722/ndss.2026.231950`

[2] D. Arp, et al., Dos and don'ts of machine learning in computer security, in: USENIX Security Symposium, 2022, pp. 3971–3988.

[3] H. Song, M. Kim, D. Park, Y. Shin, J.-G. Lee, Learning from noisy labels with deep neural networks: A survey, IEEE TNNLS 34 (11) (2022) 8135–8153.

[4] G. Patrini, A. Rozza, A. K. Menon, R. Nock, L. Qu, Making deep neural networks robust to label noise: A loss correction approach, in: CVPR, 2017, pp. 1944–1952.

[5] B. Han, et al., Co-teaching: Robust training of deep neural networks with extremely noisy labels, in: NeurIPS, 2018, pp. 8535–8545.

[6] X. Yu, et al., How does disagreement help generalization against label corruption?, in: ICML, 2019, pp. 7164–7173.

[7] E. Malach, S. Shalev-Shwartz, Decoupling "when to update" from "how to update", in: NeurIPS, 2017, pp. 960–970.

[8] J. Li, R. Socher, S. C. H. Hoi, DivideMix: Learning with noisy labels as semi-supervised learning, in: ICLR, 2020.

[9] C. G. Northcutt, L. Jiang, I. L. Chuang, Confident learning: Estimating uncertainty in dataset labels, JAIR 70 (2021) 1373–1411.

[10] T. Kim, J. Ko, J. Choi, S.-Y. Yun, FINE samples for learning with noisy labels, in: NeurIPS, 2021, pp. 24137–24149.

[11] X. Wu, W. Guo, J. Yan, B. Coskun, X. Xing, From grim reality to practical solution: Malware classification in real-world noise, in: IEEE S&P, 2023, pp. 2602–2619.

[12] Q. Yuan, G. Gou, Y. Zhu, Y. Zhu, G. Xiong, Y. Wang, MCRe: A unified framework for handling malicious traffic with noise labels, IEEE Transactions on Information Forensics and Security, 2023.

[13] T. N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: ICLR, 2017.

[14] P. Veličković, et al., Graph attention networks, in: ICLR, 2018.

[15] W. L. Hamilton, R. Ying, J. Leskovec, Inductive representation learning on large graphs, in: NeurIPS, 2017.

[16] Z. Hu, Y. Dong, K. Wang, Y. Sun, Heterogeneous graph transformer, in: The Web Conference (WWW), 2020, pp. 2704–2710.

[17] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: ICML, 2020, pp. 1597–1607.

[18] M. U. Rehman, H. Ahmadi, W. U. Hassan, Flash: A comprehensive approach to intrusion detection via provenance graph representation learning, in: IEEE S&P, 2024, pp. 3552–3570.

[19] J. Xu, X. Shu, Z. Li, Understanding and bridging the gap between unsupervised network representation learning and security analytics, in: IEEE S&P, 2024, pp. 3590–3608.

[20] L. Yang, et al., CADE: Detecting and explaining concept drift samples for security applications, in: USENIX Security Symposium, 2021, pp. 2327–2344.

[21] R. Vaarandi, A. Guerra-Manzanares, Network IDS alert classification with active learning techniques, Journal of Information Security and Applications 81 (2024) 103687.

[22] L. Chavali, A. Krishnan, P. Saxena, B. Mitra, A. Sreevallabh Chivukula, Off-policy actor-critic deep reinforcement learning for alert prioritization in intrusion detection systems, Computers & Security 142 (2024) 103854.

[23] I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani, Toward generating a new intrusion detection dataset and intrusion traffic characterization, in: ICISSP, 2018, pp. 108–116.

[24] K. Hynek, J. Luxemburk, J. Pesek, T. Cejka, P. Siska, CESNET-TLS-Year22: A year-spanning TLS network traffic dataset from backbone lines, Scientific Data 11 (2024) 1156.

[25] N. Moustafa, J. Slay, UNSW-NB15: A comprehensive data set for network intrusion detection systems, in: Military Communications and Information Systems Conference (MilCIS), 2015, pp. 1–6.

[26] J. L. Guerra, C. Catania, E. Veas, "Datasets are not enough: Challenges in labeling network traffic," *Computers & Security* 120 (2022) 102810.

[27] Y. Qing et al., "Low-quality training data only? A robust framework for detecting encrypted malicious network traffic," arXiv preprint arXiv:2309.04798, 2023.

[28] Z. Zhao et al., "Enhancing robustness of on-line learning models on highly noisy data," IEEE TDSC 18(5) (2021) 2177–2192.

[29] N. Wang et al., "FeCo: Boosting intrusion detection capability in IoT networks via contrastive learning," in: IEEE INFOCOM, 2022, pp. 1409–1418.

[30] Y. Yue et al., "Contrastive learning enhanced intrusion detection," IEEE TNSM 19(4) (2022) 4232–4247.

[31] Y. Liu et al., “DeepDRAC: Disposition recommendation for alert clusters based on security event patterns,” IEEE TIFS 20 (2025) 6443–6458.

[32] H. G. Shon, Y. Lee, M. Yoon, “Semi-supervised alert filtering for network security,” Electronics 12(23) (2023) 4755.

[33] W. Wang et al., “Transformer-based framework for alert aggregation and attack prediction in a multi-stage attack,” *Computers & Security* 136 (2024) 103533.

[34] X. Xia et al., “Sample selection with uncertainty of losses for learning with noisy labels,” arXiv preprint arXiv:2106.00445, 2021.

[35] C. Liu et al., “Noise-resistant deep metric learning with ranking-based instance selection,” in: CVPR, 2021, pp. 6811–6820.

[36] S. Yang et al., “RecDA: Concept drift adaptation with representation enhancement for network intrusion detection,” in: ACM SIGKDD, 2024, pp. 3818–3828.

[37] G. Andresini et al., “Insomnia: Towards concept-drift robustness in network intrusion detection,” in: ACM AISec, 2021, pp. 111–122.

[38] K. Hynek et al., CESNET-TLS-Year22 dataset, Zenodo record 10608607, 2024.

[39] N. Moustafa, J. Slay, “The evaluation of network anomaly detection systems: UNSW-NB15,” Information Security Journal 25(1–3) (2016) 18–31.